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NCFE · LEVEL 2 · MATHEMATICS

NCFE Functional Skills Mathematics Level 2

February 2020 Section B · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: NCFE-Maths-Level-2-Feb-2020-Section-B

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty NCFE Question Paper(s)
- Stepped MFS Multi-Colour Working & Process Marks
- Verified 100% Against Official NCFE Mark Scheme

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Activity 2: Basketball

2 (a) Lottie plays basketball.



She does some shooting practice.

She takes 75 shots.

33 of these shots go in the basket.

What percentage of her shots go in the basket?

[2 marks]

Q2(a) Worked Solution:

44(%)

Award 2 marks if correct answer given

1

33 × 100 ÷ 75

OE Any full correct method

1

44(%)

[2 marks]

Your answer: **%**

Past Paper

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2 (b) The table shows the number of points Lottie scored in Games 7 to 12 this season.

Her median score for these games is 25% higher than her median for Games 1 to 6.

Game	7	8	9	10	11	12
Score	6	10	6	14	12	10

After Game 6, Lottie's Dad said that he would increase the money in her money box by 20% each time she scored more than her median for the first six games.

Lottie started with £50 in her money box.

How much money will she have after Game 12?

[5 marks]

Q2(b) Worked Solution:

(£)103.68

Award 5 marks if correct answer given

1

10

CAO median games 7 to 12

H23a

1

Their 10 ÷ 1.25 or 8

OE any full correct method

Their 10 must be between 6 and 14,

[5 marks]

Your answer: £

Past Paper

Examiner use only

2 (c) A player's True Shooting Percentage (TS%) for a game is calculated using the following formula:

$$TS\% = \frac{50P}{0.44F + A}$$

Where: P = total points scored
F = number of free throw attempts
A = number of other goal attempts

In her last game of the season, Lottie:

- scored a total of 15 points
- attempted 8 free throws
- made 17 other goal attempts.

Use the rule to calculate Lottie's TS% for this game.

[2 marks]

Q2(c) Worked Solution:
36.5(497...) or 37 or 36 or 36.55
Award 2 marks if correct answer given
1
.
50
15
0 44
8
17
[2 marks]

Your answer:

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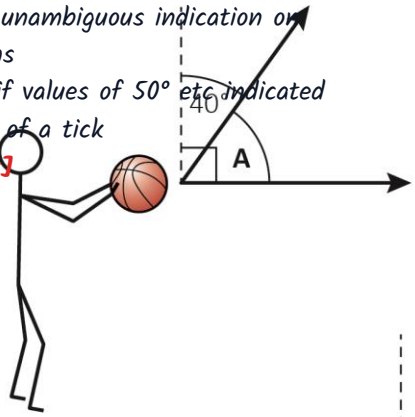
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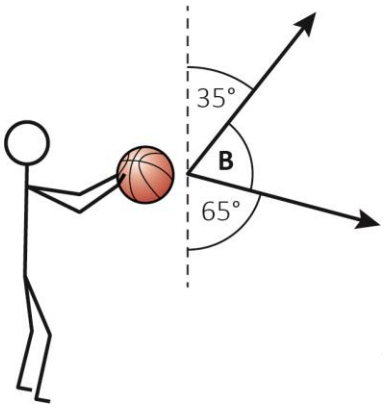
2 (d) Lottie reads that the best angle to launch the ball from for free throws is 50°

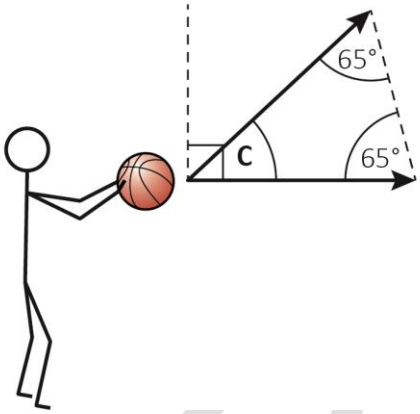
Calculate the angles A, B and C.

Q2(d) Worked Solution:

Accept unambiguous indication on diagrams
Award if values of 50° etc indicated instead of a tick
[1 mark]







Not drawn accurately

Tick the ones which are 50°

[1 mark]

Space for your workings:

Your answer:

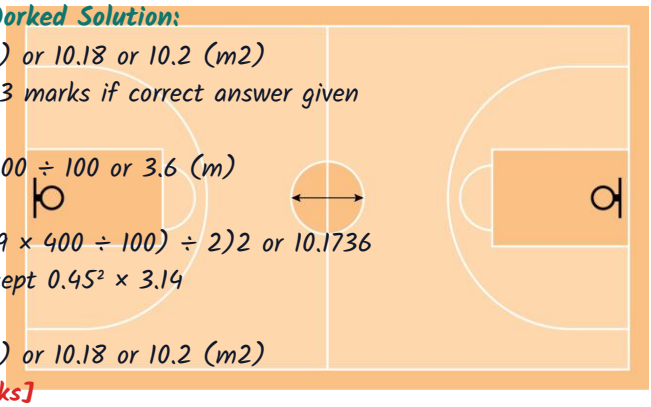
A	B	C

Past Paper

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2 (e) This is a scale drawing of a basketball court.

The scale is 1 : 400



Q2(e) Worked Solution:
10.17(...) or 10.18 or 10.2 (m2)
Award 3 marks if correct answer given
1
0.9 × 400 ÷ 100 or 3.6 (m)
1
□ ((0.9 × 400 ÷ 100) ÷ 2)2 or 10.1736
OE Accept 0.45² × 3.14
1
10.17(...) or 10.18 or 10.2 (m2)
[3 marks]

The diameter of the centre circle on the diagram is 0.9 cm

Work out the actual area of the centre circle in m^2

Use $\pi = 3.14$

[3 marks]

Your answer: **m^2**

Past Paper

Please turn over

Examiner use only

2 (f) There are 15 players in Lottie's basketball team.

The table shows information about the players:

		Height (m)		
		under 1.65 m	**1.65 to 1.8 m**	**over 1.80 m**
Usual position on court	**Centre**	0	2	2
	Forward	0	5	3
	Guard	2	1	0

Q2(f) Worked Solution:
See below
1
3
8
H26
1
0.375

The coach chooses a forward at random to take a shot.

What is the probability that the player chosen is over 1.8 m tall?

Give your answer as a fraction **and** as a decimal.

[2 marks]

FT their 3
8 if between 0 and 1 and their
decimal answer is correct to 2 or more dp
[2 marks]

Your answer:

Fraction:

Decimal:

[Total marks: 15]

Past Paper

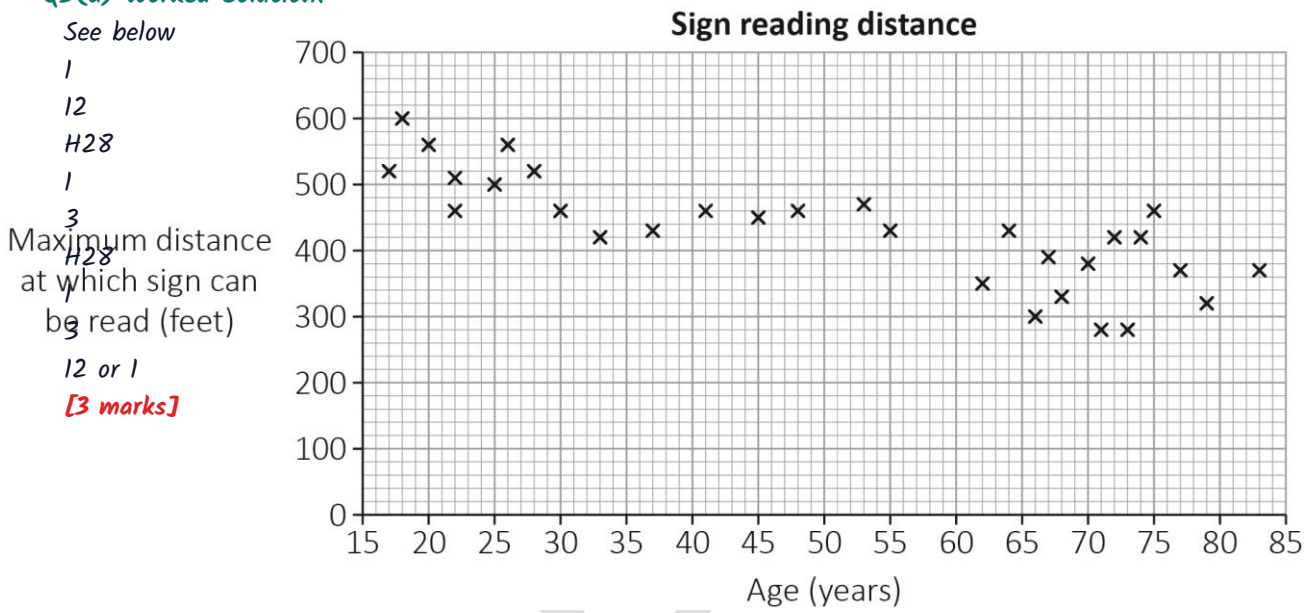
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Activity 3: Road safety

3 (a) Jed is 65 years old.
He drives a car but thinks his eyesight is not as good as it was.

The scatter diagram shows the relationship between the ages of drivers and the maximum distances at which they can read road signs.

Q3(a) Worked Solution:
See below
1
12
H28
1
3
H28
1
3
12 or 1
[3 marks]



What fraction **of the drivers over 65** can read a road sign at over 400 feet?

[3 marks]

Your answer:

Past P

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3 (b) Jed says,

"Using the scatter diagram, I predict that between the ages of 25 and 65 the maximum distance at which people can read a sign reduces by over 50 **metres"**

Is Jed's prediction reasonable?

Q3(b) Worked Solution:

Use the scatter diagram in **3 (a)** to show how you decide.

See below

Use the conversion: 1 metre = 3.28 feet

[5 marks]

Alternative 1 – working in metres

1

Scatter diagram drawn with appropriate line of best fit

Accept if in range (17, [500, 600]) and (83, [250, 350])

H28

1

[480, 560] and [340, 400]

[5 marks]

Your answer:

Please turn over

Past Paper

Examiner use only

3 (c) A Department of Transport survey collected data on the percentages of cars travelling at different speeds on a motorway.

The table shows the results of this survey:

Speed (mph)	Percentage of cars
40 ≤ speed < 50	4
50 ≤ speed < 60	13
60 ≤ speed < 70	36
70 ≤ speed < 90	47

Q3(c) Worked Solution:

Work out an estimate for the mean speed.

[2 marks]

69.95 or 70 (mph)
Award 2 marks if correct answer given
1
(45 × 4) + (55 × 13) + (65 × 36) + (80 × 47) or 6995
Allow 75 × 47 or 6760
H24
1
69.95 or 70 (mph)
H24
[2 marks]

Your answer: **mph**

Past Paper

Examiner use only

3 (d) Some drivers driving over the speed limit are invited to go to a Speed Awareness Course instead of getting penalty points.

There is a maximum speed at which a person can drive and be invited to a Speed Awareness Course.

Q3(d) Worked Solution:

42 This speed is given by this formula:

[1 mark]

$$M = 1.1L + 9$$

Where:

M is speed in miles per hour

L is the speed limit

Calculate M when L = 30 miles per hour.

[1 mark]

Your answer:

Past Paper

Please turn over

Examiner use only

3 (e) The table shows the number of people attending Speed Awareness Courses each year from 2014 to 2018:

Year	Total number of people
2014	1 355 796
2015	1 403 555
2016	1 390 880
2017	1 413 598
2018	1 445 817

Q3(e) Worked Solution:

Jed says the mean number of people attending each year for 2017 and 2018 is over 3% percent higher than the mean number for 2014 to 2016.

Is Jed right?

3.3(...) (%) or 1424912(.3) or 1424913

Show how you decide.

[4 marks]

AND yes

~~Award 4 marks if correct answer given~~

1

(1355796 + 1403555 + 1390880) ÷ 3

or (1413598 + 1445817) ÷ 2

H25

9

Level 2 – P001260 paper-based 3

1

[4 marks]

Your answer:

[Total marks: 15]

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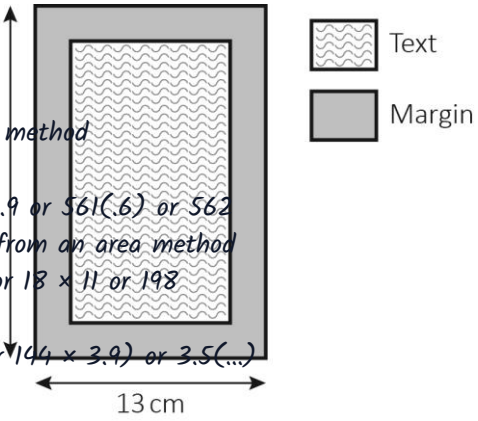
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Activity 4: Printing

4 (a) Rani is on placement at a printing company.
The company has an order for some leaflets.

Each page will have a rectangular area of text with a 2 cm margin around it.

Q4(a) Worked Solution:
See below
1
144 (cm2)
Must be area method
1
Their 144 × 3.9 or 561(.6) or 562
FT their 144 from an area method
e.g. 20 × 13 or 18 × 11 or 198
1
2000 ÷ (their 144 × 3.9) or 3.5(...)
[4 marks]



The printer can fit an average of 3.9 words of text per cm^2

There are 2000 words in total.

What is the minimum number of pages that the printer needs?

You **must** show your working.

[4 marks]

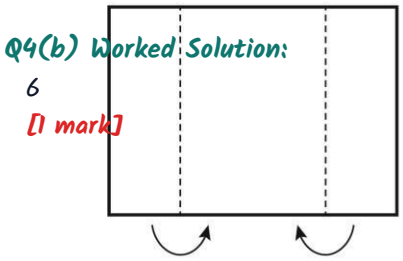
Your answer:

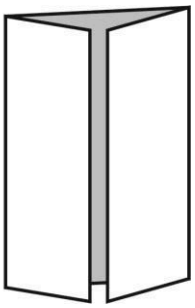
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4 (b) Another leaflet is folded like this.





What is the maximum number of printed pages the leaflet can have?

[1 mark]

Your answer:

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4 (c) The printer offers two discounts.

Offer A	Offer B
Buy 2 boxes of leaflets, get the third box half price	5% off the total price

Q4(c) Worked Solution:
See below
1
1

A customer wants to buy three boxes of leaflets.

Which offer is cheaper for the customer?

Show how you decide.

[2 marks]

6 (off)
10
Level 2 – P001260 paper-based 3
1
16(.6...) or 16.7 or 17 (%) and Offer A
Implies first mark
[2 marks]

Your answer:

Past Paper

Please turn over

Examiner use only

4 (d) Rani is asked to work out the average number of orders per month last year.

The table shows the number of orders each month.

Month	1	2	3	4	5	6	7	8	9	10	11	12
Number of orders	180	182	129	146	138	45	45	125	148	132	195	202

Q4(d) Worked Solution:

Work out both the median **and** the mode for Rani **and** give one reason why the median is the better average for her to use.

See below **[4 marks]**

I
45 45 125 129 132 138 146
or 202 195 182 180 148 146 138
H23a
I
142 (median)
Implies first mark
H23a
I
[4 marks]

Your answer:

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4 (e) The company's income in 2019 was £863 490

This was 7% higher than in 2018

Work out the company's income in 2018

[2 marks]

Q4(e) ~~Worked Solution:~~
(£)807 000
Award 2 marks if correct answer given
1
863490 ÷ 1.07
OE Any full correct method
1
(£)807 000
[2 marks]

Your answer: **£**

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4 (f) The company also prints t-shirts.

$\frac{2}{11}$ of the t-shirts are white.

$\frac{1}{3}$ of the remaining t-shirts are black.

What is the ratio of white : black t-shirts?

[2 marks]

Q4(f) Worked Solution:

2 : 3
Award 2 marks if correct answer given
1
3
11 or 2 : 9
2 : 9 is white : other
1
2 : 3
[2 marks]

Your answer:

[Total marks: 15]

This is the end of the assessment.

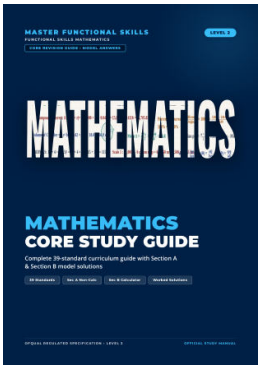
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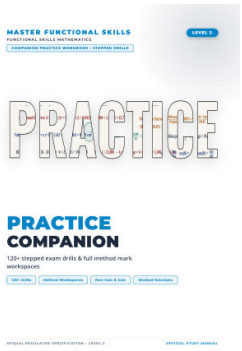


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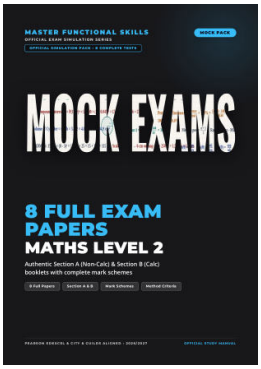


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